VERIFIED RESEARCH PORTFOLIO

Dr. Omaia Al-Omari

Evolution of cross-domain applied artificial intelligence research, 2011-2026

37 verified works	20 journal articles	13 conference papers	4 book chapters

Status and scope

This downloadable portfolio is a curated website reference edition covering 37 works verified through 24 July 2026. It is not a peer-reviewed article, an accepted manuscript, or evidence that the underlying synthesis has been published. Publication titles, years, output types, concise contributions, methods, references, DOI links, and known metadata cautions are provided for transparent academic review.

The complete source Word manuscript is intentionally not distributed through the website.

Portfolio themes

Primary theme	Works
Cybersecurity, IoT, and cloud infrastructure	9
NLP, education, and knowledge systems	8
Healthcare and clinical decision support	6
Decision analytics and intelligent systems	6
Computer vision and environmental intelligence	5
Responsible AI, governance, and society	3

Methodological trajectory

2011-2020: Classical Arabic NLP, feature engineering, statistical comparison, and connected educational systems.
2023-2024: Applied classification, multimodal attention, blockchain, reinforcement learning, and optimization.
2025: Deep learning across clinical AI, environmental intelligence, cybersecurity, governance, and decision support.
2026: Real-time detection, hybrid neural architectures, optimization, and on-demand explainable AI.

Annotated works

Records are presented chronologically. Official titles and matching APA-style references are retained in English.

P01. Evaluating the Effect of Stemming in Clustering of Arabic Documents

2011 | Journal article | NLP, education, and knowledge systems

Methods: Arabic stemming, K-means, document clustering

Al-Omari (2011) evaluated how stemming changes K-means clustering outcomes for Arabic documents, establishing morphological preprocessing as a central design choice in Arabic document clustering.

Reference: Al-Omari, O. M. (2011). Evaluating the effect of stemming in clustering of Arabic documents. Academic Research International, 1(1), 284-291.

Verification note: No DOI is available in the verified record.

P02. Arabic Part of Speech Tagging Using K-Nearest Neighbour and Naive Bayes Classifiers Combination

2014 | Journal article | NLP, education, and knowledge systems

Methods: K-nearest neighbour, Naive Bayes, majority voting, part-of-speech tagging

Mahafdah et al. (2014) combined K-nearest neighbour and Naive Bayes classifiers through majority voting and feature analysis for Quranic Arabic part-of-speech tagging, reporting a highest accuracy of 98.32%.

Reference: Mahafdah, R., Omar, N., & Al-Omari, O. (2014). Arabic part of speech tagging using K-nearest neighbour and Naive Bayes classifiers combination. Journal of Computer Science, 10(9), 1865-1873. https://doi.org/10.3844/jcssp.2014.1865.1873

P03. Ranking and Reputation Based Resource Allocation in P2P System

2017 | Journal article | Cybersecurity, IoT, and cloud infrastructure

Methods: ranking, reputation modeling, P2P resource allocation

Al Omari et al. (2017) formulated a peer-to-peer resource-allocation approach in which ranking and reputation information guide the selection and distribution of shared resources.

Reference: Al Omari, O. M. A., Khan, N. A., & Mahafdah, R. (2017). Ranking and reputation based resource allocation in P2P system. Mediterranean Journal of Basic and Applied Sciences, 1(1), 293-301.

Verification note: No DOI is available in the verified record.

P04. Learning Modal Adaptability to Improve Reading and Writing Skills of Students

2018 | Journal article | NLP, education, and knowledge systems

Methods: VARK framework, comparative learning-modal analysis

Mahafdah et al. (2018) compared six learning-modal strategies, using the VARK framework to examine how modality preferences and gender-related patterns may inform support for students' reading and writing development.

Reference: Mahafdah, R. F., Al-Omari, O. M., & Khan, N. A. (2018). Learning modal adaptability to improve reading and writing skills of students. International Journal of Engineering Research & Technology, 7(3). https://doi.org/10.17577/IJERTV7IS030016

P05. Enhanced Document Classification Using Noun Verb (NV) Terms Extraction Approach

2019 | Journal article | NLP, education, and knowledge systems

Methods: noun-verb term extraction, KNN, Naive Bayes, SVM

Al-Omari and Omari (2019) showed that jointly extracting nouns and verbs can strengthen document classification across KNN, Naive Bayes, and SVM models compared with noun-only, verb-only, and bag-of-words representations.

Reference: Al-Omari, O., & Omari, N. (2019). Enhanced document classification using noun verb (NV) terms extraction approach. International Journal of Advanced Trends in Computer Science and Engineering, 8(1), 85-92.
https://doi.org/10.30534/ijatcse/2019/15812019

Verification note: the publisher/author record assigns this DOI to the article above, but the current Crossref metadata resolves the same DOI to a different title. The inconsistency must be checked with the publisher before journal submission.

P06. Prevention of Web-Form Spamming for Cloud-Based Applications: A Proposed Model

2019 | Conference paper | Cybersecurity, IoT, and cloud infrastructure

Methods: AI-assisted filtering, application-layer security, cloud applications

Khan et al. (2019a) proposed an artificial-intelligence-assisted application-layer model for filtering malicious web-form requests in cloud applications before services are granted.

Reference: Khan, N. A., Siddiqi, A. M. U., Mohammad, M. A., Ahmad, D., Hameed, S. A., & Al Omari, O. M. (2019a). Prevention of web-form spamming for cloud-based applications: A proposed model. In 2019 Amity International Conference on Artificial Intelligence (AICAI) (pp. 249-254). IEEE.
https://doi.org/10.1109/AICAI.2019.8701302

P07. Intrusion Management to Avoid Web-Form Spamming in Cloud-Based Architectures

2019 | Conference paper | Cybersecurity, IoT, and cloud infrastructure

Methods: intrusion management, application-layer filtering, cloud security

Khan et al. (2019b) designed an application-layer intrusion-management mechanism intended to identify and block web-form spam within cloud-based architectures.

Reference: Khan, N. A., Mahafdah, R. F., Nasimuddin, M. A. M., Alomari, O. M., Siddiqi, A. M. U., & Ahamad, D. (2019b). Intrusion management to avoid web-form spamming in cloud-based architectures. In 2019 International Conference on Computational Intelligence and Knowledge Economy (ICCIKE) (pp. 437-442). IEEE.
https://doi.org/10.1109/ICCIKE47802.2019.9004302

P08. Analytical Assessment of Noun Verb Term Extraction for Document Classification Using T-Test

2020 | Journal article | NLP, education, and knowledge systems

Methods: noun-verb term extraction, KNN, Naive Bayes, SVM, t-test

Al-Omari (2020a) statistically assessed noun-verb term extraction across KNN, Naive Bayes, and SVM classifiers using benchmark document collections and t-test-based comparisons.

Reference: Al-Omari, O. M. (2020a). Analytical assessment of noun verb term extraction for document classification using t-test. Journal of Mechanics of Continua and Mathematical Sciences, 15(3).
https://doi.org/10.26782/jmcms.2020.03.00021

P09. Internet of Things (IoT)-Based Educational Data Mining (EDM) System

2020 | Journal article | NLP, education, and knowledge systems

Methods: Internet of Things, RFID, educational data mining

Al-Omari (2020b) proposed an IoT- and RFID-enabled educational data-mining model that connects student engagement records, university analytics, and industry-institute linkage.

Reference: Al-Omari, O. M. (2020b). Internet of Things (IoT)-based educational data mining (EDM) system. Journal of Mechanics of Continua and Mathematical Sciences, 15(3).
https://doi.org/10.26782/jmcms.2020.03.00022

P10. A Review Methodology for COVID-19 Using Enhanced Techniques

2020 | Journal article | Healthcare and clinical decision support

Methods: structured review, COVID-19 analytics

Nagaraju et al. (2020) synthesized the structure, clinical context, and emerging analytical approaches associated with COVID-19 to frame directions for technology-supported review and assessment.

Reference: Nagaraju, R., Alahmari, S. A., Patra, P. S. K., Rajeyyagari, S., & Al-Omari, O. M. (2020). A review methodology for COVID-19 using enhanced techniques. Journal of Critical Reviews, 7(18), 2116-2122.
https://doi.org/10.31838/jcr.07.18.264

P11. An Empirical Study on the Future of Publication Repositories and Its Adaptability in Public Universities

2023 | Book chapter | NLP, education, and knowledge systems

Methods: case study, repository adoption, adaptability assessment

Khan et al. (2023) examined the acceptability and adaptability of institutional publication repositories through a Shaqra University case study based on data collected over two months.

Reference: Khan, N. A., Al-Omari, O. M., & Alshahrani, S. M. (2023). An empirical study on the future of publication repositories and its adaptability in public universities: A case study of Shaqra University, Saudi Arabia. In Computational Intelligence (pp. 823-829). Springer. https://doi.org/10.1007/978-981-19-7346-8_71

P12. IoT and Machine Learning Models to Develop the Architecture for Assessment of Security Threats

2023 | Conference paper | Cybersecurity, IoT, and cloud infrastructure

Methods: IoT architecture review, machine-learning threat assessment

Alotibi et al. (2023) reviewed IoT architecture, threat classes, and machine-learning-based security mechanisms to frame an architecture for assessing cyber-physical security threats.

Reference: Alotibi, M. M., Rajeyyagari, S., Al-Omari, O. M., & Alymani, M. (2023). IoT and machine learning models to develop the architecture for assessment of security threats. In 2023 International Conference on New Frontiers in Communication, Automation, Management and Security (ICCAMS) (pp. 1-6). IEEE. https://doi.org/10.1109/ICCAMS60113.2023.10526005

P13. MuSeRAN: Multimodal Sentiment Analysis with Recurrent Attention Networks

2024 | Conference paper | Decision analytics and intelligent systems

Methods: recurrent attention networks, multimodal sentiment analysis

Al-Omari et al. (2024) introduced a recurrent-attention architecture for combining multiple modalities in aspect-oriented sentiment analysis, extending sentiment modeling beyond single-channel textual evidence.

Reference: Al-Omari, O., Omari, A., Othman, E., & Alrashed, A. A. (2024). MuSeRAN: Multimodal sentiment analysis with recurrent attention networks. In 2024 International Conference on Decision Aid Sciences and Applications (DASA) (pp. 1-5). IEEE. https://doi.org/10.1109/DASA63652.2024.10836479

P14. Combination of Blockchain and Machine Learning for Secure Supply Chain Authentication Management

2024 | Conference paper | Decision analytics and intelligent systems

Methods: blockchain, machine learning, supply-chain analytics

Allahham et al. (2024) proposed integrating blockchain's immutable transaction record with machine-learning analytics to strengthen supply-chain authentication, forecasting, and decision support.

Reference: Allahham, M., Alrashed, A. A., Al-Omari, O., & Almajali, W. I. (2024). Combination of blockchain and machine learning for the improvement of decision making and secure supply chain authentication management. In 2024 International Conference on Decision Aid Sciences and Applications (DASA) (pp. 1-4). IEEE.
https://doi.org/10.1109/DASA63652.2024.10836579

P15. Enhancing Feedback Mechanisms in English Language Assessments Through Deep Learning Model

2024 | Conference paper | NLP, education, and knowledge systems

Methods: Double Integral Transformer, local-global attention, language assessment

Selvi et al. (2024) proposed a Double Integral Transformer that combines local and global contextual attention to produce more targeted feedback in English-language assessment.

Reference: Selvi, M. D. T., Zeeshan, G. A., Al-Omari, O., Mishra, A. K., Dorcas, G. E., & Alphonse, F. R. (2024). Enhancing feedback mechanisms in English language assessments through deep learning model. In 2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Application (ICAIQSA) (pp. 1-5). IEEE.
https://doi.org/10.1109/ICAIQSA64000.2024.10882425

P16. Forecasting Stock Market Using Regression Models

2024 | Conference paper | Decision analytics and intelligent systems

Methods: regression models, random-forest regression, market forecasting

Omari et al. (2024) compared regression-based forecasting approaches on historical market data and identified random-forest regression as the strongest evaluated model without evidence of overfitting in the reported experiment.

Reference: Omari, A., Alazzam, A., & Al-Omari, O. (2024). Forecasting stock market using regression models. In Proceedings of the 10th International Conference on Industrial and Business Engineering (pp. 111-118). ACM.
https://doi.org/10.1145/3716097.3716124

P17. Smart Cloud Data Management Based on Deep Reinforcement Learning with Spider Swarm Optimization

2024 | Book chapter | Cybersecurity, IoT, and cloud infrastructure

Methods: deep reinforcement learning, Spider Swarm Optimization, cloud data management

Nallola et al. (2024) combined deep reinforcement learning with spider-swarm optimization to frame an adaptive approach for intelligent cloud data management.

Reference: Nallola, S. R., Al-Omari, O. M., Ayyasamy, V., & Inbavalli, A. (2024). Smart cloud data management based on deep reinforcement learning with spider swarm optimization algorithm. In Advancing Sustainable Science and Technology for a Resilient Future (pp. 11-15). CRC Press. https://doi.org/10.1201/9781003490210-3

P18. Enhancement of Cyber Security in IoT Based on Ant Colony Optimized Artificial Neural Adaptive TensorFlow

2024 | Journal article | Cybersecurity, IoT, and cloud infrastructure

Methods: Ant Colony Optimization, artificial neural networks, recurrent neural network, TensorFlow

Sadu et al. (2024) combined ant-colony optimization, neural learning, and a multi-objective recurrent network to detect malicious software and suspicious activity in IoT environments.

Reference: Sadu, V. B., Abhishek, K., Al-Omari, O. M., Nallola, S. R., Sharma, R. K., & Khan, M. S. (2024). Enhancement of cyber security in IoT based on ant colony optimized artificial neural adaptive TensorFlow. Network: Computation in Neural Systems, 36(3), 598-614. https://doi.org/10.1080/0954898X.2024.2336058

P19. AI-Powered Climate Change Analytics Approach

2025 | Conference paper | Computer vision and environmental intelligence

Methods: SARIMA, decision tree, time-series forecasting, climate analytics

Omari et al. (2025) used SARIMA forecasting and decision-tree classification on Berkeley Earth temperature data, identifying a significant warming trend while documenting the limits of region-level classification from restricted features.

Reference: Omari, A., Alazzam, A., Badreddine, S., & Al-Omari, O. (2025). AI-powered climate change analytics approach. In 2025 2nd International Conference on Artificial Intelligence, Metaverse, and Cybersecurity (ICAMAC) (pp. 1-5). IEEE. https://doi.org/10.1109/ICAMAC67779.2025.11398634

P20. Deep Learning-Based Automated Segmentation of the Parcellated Corpus Callosum in Brain MRI

2025 | Journal article | Healthcare and clinical decision support

Methods: attention U-Net, residual blocks, MRI segmentation

Fati and Al-Omari (2025) integrated residual blocks and attention gates into U-Net for five-region corpus-callosum segmentation, reporting Dice scores near 97% across ABIDE, OASIS, and clinical-image datasets.

Reference: Fati, S. M., & Al-Omari, O. (2025). Deep learning-based automated segmentation of the parcellated corpus callosum in brain MRI. Engineering, Technology & Applied Science Research, 15(5), 27357-27362. https://doi.org/10.48084/etasr.11783

P21. CNN-Based Automated Detection of Metastatic Cancer in Histopathology Images

2025 | Journal article | Healthcare and clinical decision support

Methods: EfficientNetB3, Grad-CAM, histopathology classification

Al-Omari et al. (2025a) used EfficientNetB3 and Grad-CAM on BreaKHis histopathology images, reporting 92.33% test accuracy while localizing diagnostically relevant regions.

Reference: Al-Omari, O., Alkhatib, O., & Al-Omari, T. (2025a). CNN-based automated detection of metastatic cancer in histopathology images. Engineering, Technology & Applied Science Research, 15(4), 24478-24485.
https://doi.org/10.48084/etasr.10888

P22. Investigating Brain-Inspired Computing Architectures to Enhance Energy Efficiency in AI Models

2025 | Conference paper | Decision analytics and intelligent systems

Methods: deep neural networks, neuromorphic computing, energy-efficient AI

Bai et al. (2025) proposed a hybrid deep-neural and neuromorphic architecture that shifts energy-constrained inference toward event-driven processing while preserving competitive predictive performance.

Reference: Bai, M. R., Priyanka, Al-Omari, O., Devi, N., Dhas, J. T. M., & Bala, B. K. (2025). Investigating brain-inspired computing architectures to enhance energy efficiency in AI models using neuromorphic computing. In 2025 International Conference on Innovations in Intelligent Systems: Advancements in Computing, Communication, and Cybersecurity (ISAC3) (pp. 1-6). IEEE.
https://doi.org/10.1109/ISAC364032.2025.11156604

P23. Artificial Intelligence and Posthumanism: A Philosophical Inquiry into Consciousness, Ethics, and Human Identity

2025 | Journal article | Responsible AI, governance, and society

Methods: philosophical inquiry, posthumanism, AI ethics

Al-Omari and Al-Omari (2025) situated artificial intelligence within posthumanist debate, examining how increasingly capable systems challenge assumptions about consciousness, moral status, human uniqueness, and identity.

Reference: Al-Omari, O., & Al-Omari, T. (2025). Artificial intelligence and posthumanism: A philosophical inquiry into consciousness, ethics, and human identity. Journal of Posthumanism, 5(2), 458-469.
https://doi.org/10.63332/joph.v5i2.432

P24. Object Detection in Real-Time Video Surveillance Using Attention-Based Transformer-YOLOv8 Model

2025 | Journal article | Computer vision and environmental intelligence

Methods: YOLOv8, Transformer, attention, real-time object detection

Nimma et al. (2025) combined attention, transformer context modeling, and YOLOv8 for real-time surveillance, reporting 96.78% precision, 96.89% recall, and 89.67% mean average precision.

Reference: Nimma, D., Al-Omari, O., Pradhan, R., Ulmas, Z., Krishna, R. V. V., El-Ebiary, Y. A. B., & Rao, V. S. (2025). Object detection in real-time video surveillance using attention-based transformer-YOLOv8 model. Alexandria Engineering Journal, 118, 482-495.
https://doi.org/10.1016/j.aej.2025.01.032

P25. Governance and Ethical Frameworks for AI Integration in Higher Education

2025 | Journal article | Responsible AI, governance, and society

Methods: AI governance framework, ethical analysis, legal compliance

Al-Omari et al. (2025b) developed a governance-oriented account of higher-education AI that links personalized learning with privacy, bias control, legal compliance, capacity building, and institutional accountability.

Reference: Al-Omari, O., Alyousef, A., Fati, S., Shannaq, F., & Omari, A. (2025b). Governance and ethical frameworks for AI integration in higher education: Enhancing personalized learning and legal compliance. Journal of Ecohumanism, 4(2).
https://doi.org/10.62754/joe.v4i2.5781

P26. Artificial Intelligence in Healthcare: Bridging Innovation and Regulation

2025 | Journal article | Responsible AI, governance, and society

Methods: healthcare AI regulation, safety and accountability analysis

Alyousef and Al-Omari (2025) connected diagnostic and personalized-medicine opportunities with the need for clear safety, accountability, and regulatory requirements for clinical AI.

Reference: Alyousef, A., & Al-Omari, O. (2025). Artificial intelligence in healthcare: Bridging innovation and regulation. Journal of Ecohumanism, 3(8).
https://doi.org/10.62754/joe.v3i8.5673

P27. A Deep Learning Approach for Proactive Detection and Mitigation of Zero-Day Attacks in IoT Environment

2025 | Book chapter | Cybersecurity, IoT, and cloud infrastructure

Methods: deep learning, anomaly detection, zero-day attack mitigation

Alijoyo et al. (2025) framed zero-day defense in IoT as a proactive deep-learning problem that couples early anomaly detection with mitigation actions rather than relying only on known attack signatures.

Reference: Alijoyo, F. A., Venkatramana, N., Al-Omari, O., Khan, S. A., & Bala, B. K. (2025). A deep learning approach for proactive detection and mitigation of zero-day attacks in IoT environment. In Lecture Notes in Networks and Systems (pp. 47-61). Springer.
https://doi.org/10.1007/978-981-96-6429-0_5

P28. A Machine Learning Approach to Analyze the Shift from In-Person to Virtual Care for Mental Health Services

2025 | Conference paper | Healthcare and clinical decision support

Methods: machine learning, billing-data analysis, virtual-care analytics

Omari et al. (2025) analyzed billing data from April 2019 to March 2021 to characterize the shift between in-person and virtual mental-health services across family medicine, pediatrics, and psychiatry.

Reference: Omari, A., Al-Omari, O., & Badreddine, S. (2025). A machine learning approach to analyze the shift from in-person to virtual care for mental health services. In 2025 14th Mediterranean Conference on Embedded Computing (MECO) (pp. 1-5). IEEE.
https://doi.org/10.1109/MECO66322.2025.11049284

P29. A Predictive Analytics Approach to Improve Telecom's Customer Retention

2025 | Journal article | Decision analytics and intelligent systems

Methods: SVM, feature selection, churn prediction, model interpretation

Omari et al. (2025) integrated preprocessing, feature selection, and model interpretation for telecom churn prediction, with the support-vector machine providing the strongest performance among the evaluated models.

Reference: Omari, A., Al-Omari, O., Al-Omari, T., & Fati, S. M. (2025). A predictive analytics approach to improve telecom's customer retention. Frontiers in Artificial Intelligence, 8, 1600357.
https://doi.org/10.3389/frai.2025.1600357

P30. Applying Artificial Intelligence Techniques on Cyber Attacks Dataset

2025 | Book chapter | Cybersecurity, IoT, and cloud infrastructure

Methods: AI classification, cyberattack dataset analysis

Al-Omari et al. (2025c) evaluated selected artificial-intelligence techniques on cyberattack data to support data-driven attack classification and detection.

Reference: Al-Omari, O., AlShboul, Y., Alyousef, A., Shannaq, F., & Omari, A. (2025c). Applying artificial intelligence techniques on cyber attacks dataset. In Lecture Notes on Data Engineering and Communications Technologies (pp. 74-84). Springer.
https://doi.org/10.1007/978-3-031-87778-0_7

P31. Challenges and Solutions in Agile Software Development: A Managerial Perspective

2025 | Journal article | Decision analytics and intelligent systems

Methods: managerial synthesis, Agile adoption analysis

Geetha et al. (2025) synthesized managerial barriers to Agile adoption - including resistance, limited expertise, coordination problems, and uneven stakeholder commitment - and mapped them to training, leadership, phased transition, and communication interventions.

Reference: Geetha, L. S., El-Ebiary, Y. A. B., Rao, B. S., Rautrao, R. R., Rao, T. S. M., Ramesh, J. V. N., & Al-Omari, O. (2025). Challenges and solutions in Agile software development: A managerial perspective on implementation practices. International Journal of Advanced Computer Science and Applications, 16(3). https://doi.org/10.14569/IJACSA.2025.0160374

P32. LSTM Networks for Improved Environmental Monitoring and Biodiversity Conservation

2025 | Conference paper | Computer vision and environmental intelligence

Methods: LSTM, environmental time-series modeling, biodiversity monitoring

Raju et al. (2025) applied LSTM modeling to environmental time series and biodiversity dynamics, reporting 97.6% accuracy for the evaluated monitoring task and emphasizing temporal prediction for conservation decisions.

Reference: Raju, D., Al-Omari, O., Anbazhagan, G. K., Selladurai, K. M., Alazzam, M. B., & Karthik, M. (2025). LSTM networks for improved environmental monitoring and biodiversity conservation. In 2025 International Conference on Emerging Smart Computing and Informatics (ESCI) (pp. 1-5). IEEE. https://doi.org/10.1109/ESCI63694.2025.10988428

P33. LiDAR-Based Climate Change Imaging in Geoscience Using Spatio Extreme Fuzzy Gradient Model

2025 | Journal article | Computer vision and environmental intelligence

Methods: LiDAR, spatio-extreme fuzzy gradient, remote sensing

Krushnasamy et al. (2025) combined LiDAR-based geoscience imaging with a spatio-extreme fuzzy-gradient model to represent climate-related spatial variation in remote-sensing data.

Reference: Krushnasamy, V. S., Al-Omari, O., Sundaram, A., Varghese, I. K., Muniyandy, E., & Rao, M. V. A. L. N. (2025). LiDAR-based climate change imaging in geoscience using spatio extreme fuzzy gradient model. Remote Sensing in Earth Systems Sciences, 8(2), 465-472. https://doi.org/10.1007/s41976-025-00197-5

P34. KAN-ALSTM: A Clinical Decision Aid for Predicting Parkinson's Disease Progression

2025 | Conference paper | Healthcare and clinical decision support

Methods: Kolmogorov-Arnold Networks, attention LSTM, multimodal fusion, SHAP

Al-Omari et al. (2025d) combined Kolmogorov-Arnold Networks, attention-based LSTMs, multimodal fusion, and SHAP explanations to model Parkinson's disease progression from heterogeneous clinical and sequential voice data.

Reference: Al-Omari, O., Alzoubi, M. Y., Alyousef, A., Othman, E., Omari, A., & Alrashed, A. A. (2025d). KAN-ALSTM: A clinical decision aid for predicting Parkinson's disease progression via Kolmogorov-Arnold Networks and attention-based LSTMs. In 2025 International Conference on Decision Aid Sciences and Applications (DASA) (pp. 2190-2194). IEEE. https://doi.org/10.1109/DASA68193.2025.11498856

P35. XAI-Driven Decision Aid for Chronic Disease Progression

2025 | Conference paper | Healthcare and clinical decision support

Methods: gradient boosting, SHAP, clinical forecasting

Al-Omari et al. (2025e) paired gradient boosting with SHAP-based attribution to frame a transparent clinical forecasting approach for chronic-disease progression.

Reference: Al-Omari, O., AlShboul, Y., Mahafdah, R., Alzoubi, M. Y., Omari, A., & Alrashed, A. A. (2025e). XAI-driven decision aid for chronic disease progression: Combining gradient boosting and SHAP for transparent clinical forecasting. In 2025 International Conference on Decision Aid Sciences and Applications (DASA) (pp. 1015-1019). IEEE. https://doi.org/10.1109/DASA68193.2025.11498906

P36. Towards Trustworthy Defense AI: Real-Time Military Asset Detection with On-Demand Explainable YOLOv8

2026 | Journal article | Computer vision and environmental intelligence

Methods: YOLOv8l, SHAP, on-demand XAI, real-time detection

Al-Omari et al. (2026) integrated YOLOv8l with on-demand SHAP explanations for twelve classes of military assets, reporting 92.3% mAP@0.5 and 35 frames per second while reserving explainability for high-risk detections or operator requests.

Reference: Al-Omari, O., Alyousef, A., Fati, S. M., Othman, E., & Naeem, M. R. (2026). Towards trustworthy defense AI: Real-time military asset detection with on-demand explainable YOLOv8. Engineering, Technology & Applied Science Research, 16(2), 33971-33978. https://doi.org/10.48084/etasr.17259

P37. Phishing Detection Using CNN-RNN with Sparrow Search Optimization Algorithm

2026 | Journal article | Cybersecurity, IoT, and cloud infrastructure

Methods: CNN-RNN, Sparrow Search Optimization, phishing detection

Alshboul et al. (2026) combined CNN-RNN feature learning with Sparrow Search optimization for website and email phishing detection, reporting 99.27% accuracy and a 99.12% F1-score.

Reference: Alshboul, Y., Al-Omari, O., & Bsoul, A. (2026). Phishing detection using CNN-RNN with Sparrow Search optimization algorithm. Journal of Multiscale Modelling, 17(1), 2640021. https://doi.org/10.1142/S1756973726400214

Verification notes

P01 and P03 have no DOI in the verified record. P05 is retained with a metadata warning because the publisher/author record and current Crossref resolution conflict. Final publication-year confirmation remains advisable for online-first records before journal submission or formal bibliometric analysis.

For current citation counts and the author's live publication profile, consult Google Scholar and ORCID separately; citation metrics are intentionally excluded from this fixed portfolio.

End of portfolio